Agnosia

Martha J. Farah

Carnegie Mellon University, Pittsburgh, Pennsylvania, USA

Object recognition can break down in a variety of ways after brain damage. The resulting different forms of agnosia provide us with useful constraints on theories of normal object recognition. Recent studies suggest a division of labor for the recognition of different types of stimuli (common objects, words, faces, direction of eye gaze, spatial relations among parts of the human body), a high degree of interactivity in the processes underlying object recognition, and the possibility that recognition and awareness of recognition may be neurally distinct.

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Introduction

The term agnosia covers a multitude of disorders of high-level perception that follow brain damage. The definition of agnosia is largely exclusionary: impairment of object recognition not caused by sensory deficits or generalized intellectual loss. Within these broad definitional boundaries there are many distinct disorders. Indeed, much of the relevance of agnosia to cognitive neuroscience stems from this heterogeneity. The variety of different ways in which perception can break down following brain damage provides evidence that it is a complex process with multiple isolable subsystems. The question of how best to parse the array of different high-level perceptual impairments in order to interpret them in terms of theories of normal perception, and use them as evidence for testing theories of perception, has been the topic of recent discussion in cognitive neuropsychology [1,2].

Within cognitive neuroscience, agnosia has been studied with the goal of elucidating the 'functional architecture' of high-level perception, that is, delineating the different, relatively independent, information-processing subsystems that underlie perception, and characterizing their functions and interactions. In this review I will focus on three issues related to this general goal. First, are there different specialized subsystems for recognizing different types of objects and if so, what are they? Second, how interactive is the visual information processing that underlies object recognition? Third, can visual recognition be dissociated from awareness of recognition, and what are the theoretical implications of such a dissociation? Except where noted, the work reviewed here all concerns visual perception, reflecting the general bias of the field.

Specialization within the object recognition system

Selective visual agnosias: face, object and printed word recognition

Within disorders of visual recognition, there are numerous cases of pairwise dissociations among face recognition, object recognition, and printed word recognition. This suggests that there is some division of labor for different types of stimuli within the visual recognition system. A meta-analysis of 99 such cases revealed that not all combinations of spared and impaired face, object and word recognition occur, and that the pattern of association and dissociation among these abilities is consistent with the existence of two types of visual pattern recognition processes: one that is essential for face recognition, used for object recognition, and not needed for reading; and another that is essential for reading, used for object recognition, and not needed for face recognition [3]. The former appears to depend on ventral temporo–occipital regions bilaterally, but may occasionally be compromised after a unilateral right hemisphere lesion; the latter appears to depend on left temporo–occipital cortex.

Much of the recent work on impaired face recognition, known as prosopagnosia, has focused on unconscious face recognition and is discussed in the third section of this review. A recent study of eye gaze perception in prosopagnosia deserves mention here, however, because of its surprising implications for the functional architecture of high-level vision [4•]. It has been found that prosopagnosic patients are impaired at discriminating whether someone is looking directly at them, or to one side. They are not impaired at similar visual–spatial discrimination tasks not involving faces. This suggests that the perception of gaze direction is not handled by general purpose visual mechanisms. Rather, it is carried out by a specialized subsystem of the functional architecture of vision, which evidently evolved to perform face-related perception.

Research on impairments of printed word recognition, known as the alexias, has recently focused on the role of perceptual factors in reading disorders. A special double issue of the journal *Cognitive Neuropsychology* [5•] was devoted to the different forms of alexia that can arise from visual disorders, including general disorders of visual attention and of visual pattern recognition. If the alexias can be explained by loss of general-purpose visual and/or linguistic abilities, rather than reading-specific abilities, then we do not need to posit reading-specific components of the functional architecture of vision. This represents an important theoretical departure from earlier interpretations of alexia in terms of loss of 'visual word-form' systems, but seems consistent with the late development, phylogenetically and ontogenetically, of reading as compared with, for example, face recognition.

Non-visual agnosias

Two recent studies of tactile agnosia [6,7] have greatly expanded our knowledge of this syndrome, demonstrating the existence of tactile object recognition impairments in the absence of visual recognition impairments and significant sensory or motor impairment. This is consistent with object recognition in the visual and tactile modalities relying on separate components of the functional architecture.

Another recent contribution to non-visual agnosia is a striking case study of autotopagnosia [8••], also known as 'loss of body schema'. The patient in this study was severely impaired at localizing named or depicted body parts on herself, the examiners, or a doll. For example, she could not point to her elbow when asked. That the disorder was specific to the spatial structure of the body was demonstrated by contrasting her body part localization with the localizations of other small objects. When the objects were pinned to the patient's body, she easily localized them. Even when removed, she was able to localize their former positions, for example, pointing to her elbow when a particular object was named, if that was the former location of the object. This suggests that there is a surprisingly strict division of labor within the functional architecture of perception, according to which the spatial lesions among body parts and the spatial relations among other items are represented by different systems. Body perception appears not to be purely visual, as the patient was impaired at pointing to parts of her own body with her eyes closed.

Interactivity among the components of object recognition

What are the relations among different components of the functional architecture? How interactive are the different hypothesized components of perceptual processing? Does each component carry out its computation and then pass the result to the next one, or is the entire process of computation carried out in a continuous interactive 'give and take' with other components? Three recent studies suggest that interactivity may be the rule in visual recognition.

A patient with simultanagnosia, an impairment in the ability to perceive more than one object at a time, was tested with multiple items that were either semantically related to one another (e.g. shirt and pants) or unrelated [9••]. When related items were shown, he was able to perceive more of them. This suggests that the perception of one object activated high-level knowledge representations, and that these in turn influenced the perception of a second object, partially circumventing the visual processing limitations of simultanagnosia.

Patients with prosopagnosia are found to be relatively preserved in their ability to discriminate simple components of face patterns (e.g. the shape of a mouth), consistent with intact low-level visual perception. Unlike normal subjects, however, whose perception of pattern components is facilitated by the presence of the full intact pattern context, the prosopagnosics were not influenced by the context [10]. This is consistent with the interactive nature of perception — whereas the simultanagnosic's impaired vision can be facilitated by higher-level representations because these are intact, prosopagnosics' low-level vision cannot be facilitated by higher-level visual representations because these are impaired.

A recent computer simulation of visual word recognition shows how extensive interaction among components of the functional architecture, such as visual pattern analysis and semantic knowledge, can accomplish the task of word recognition, and how damage to any part of the system causes errors that reflect the functioning of all parts [11••]. For example, a single lesion to the semantic component of the model creates both semantic errors (e.g. reading 'apple' as 'orange') and visual errors (e.g. reading 'house' as 'horse').

The relation between recognition and awareness

What is the relation between perception and awareness of perception. Do recognition and awareness of recognition rely on the same, or partially distinct, components of the functional architecture?

Covert recognition in prosopagnosia

Some prosopagnosic patients can manifest recognition of faces when tested in certain indirect ways (for a recent review, see [12]). For example, their speed of classifying printed names as politicians or actors is affected by whether or not a simultaneously presented face belongs in the same category as the name. This has been taken to imply that face recognition *per se* is not impaired in these patients, but rather that there is a disconnection between the face recognition system and other systems necessary for conscious awareness. This in turn implies that recognition and awareness of recognition depend on different components of the functional architecture.

A version of this hypothesis has recently been implemented in a computer simulation model [13•], and shown to account for findings of priming and interference by faces, as in the 'actor/politician' example given above.

Recognition for action and awareness

A study of a patient with apperceptive agnosia, that is, impaired object recognition secondary to impaired low-level shape perception, also shows a dissociation between recognition tested in different ways [14••]. Although the patient performed poorly at naming or discriminating shapes, she showed relatively preserved shape perception, as measured by her actions towards objects (e.g. reaching for objects with the appropriate hand shape). The authors' interpretation of this dissociation is analogous to that proposed for covert recognition in prosopagnosia; shape representations may have been successfully computed, but are available only to the motor system, and not to other systems necessary for conscious awareness. In both cases this implies that there is a difference between the brain systems necessary for computing particular stimulus properties (facial identity or object shape), and those necessary for a subject to be consciously aware of these aspects of the stimulus. Whether or not this is the correct inference from these data remains to be determined (see [1]).

Conclusions

For much of its history, the study of agnosia has been primarily descriptive. The different forms of agnosia are sufficiently numerous, and pure cases are sufficiently rare, that the mere documentation of the disorders has occupied researchers in this area. More recently, cognitive neuropsychologists have shifted from studying agnosia as an end in itself, to studying it as a means of understanding normal object recognition. The articles reviewed above attest to the fruitfulness of this approach. The different patterns of spared and impaired recognition abilities, the mutual influence of different levels of perception and knowledge in visual recognition, and the existence of recognition without awareness, all contribute useful constraints on theories of normal object recognition.

References and recommended reading

Papers of particular interest, published within the annual period of review, have been highlighted as:
• of special interest
•• of outstanding interest

1. FARAH MJ: *Visual Agnosia: Disorders of Object Recognition and What They Tell Us About Normal Vision.* Cambridge: MIT Press; 1990.

2. HUMPHREYS GW, RIDDOCH MJ: *To See But Not To See.* Hillsdale, New Jersey: Lawrence Erlbaum; 1987.

3. FARAH MJ: **Patterns of Co-occurrence Among the Associative Agnosias: Implications for Visual Object Representations.** *Cogn Neuropsychol* 1991, **8**:1–19.

4. • CAMPBELL R, HEYWOOD CA, COWEY A, REGARD M, LANDIS T: **Sensitivity to Eye Gaze in Prosopagnosic Patients and Monkeys with Superior Temporal Sulcus Ablation.** *Neuropsychologia* 1990, **28**:1123–1142.

Converging evidence from human prosopagnosics and monkeys with lesions of superior temporal sulcus showing specialized mechanisms for the perception of eye gaze direction.

5. • RIDDOCH MJ: **Special Issue on Neglect and the Peripheral Dyslexias.** *Cogn Neuropsychol* 1990, **7**.

A collection of case studies testing various hypotheses concerning the relations between perception, attention and reading.

6. CASELLI RJ: **Rediscovering Tactile Agnosia.** *Mayo Clin Proc* 1991, **66**:129–142.

7. CASELLI RJ: **Bilateral Impairment of Somesthetically Mediated Object Recognition in Humans.** *Mayo Clin Proc* 1991, **66**:357–364.

8. •• SIRIGU A, GRAFMAN J, BRESSLER K, SUNDERLAND T: **Multiple Representations Contribute to Body Knowledge Processing. Evidence from a Case of Autotopagnosia.** *Brain* 1991, **114**:629–642.

A strikingly selective impairment of 'body schema' documented with ingenuity and thoroughness.

9. •• COSLETT HB, SAFFRAN E: **Simultanagnosia. To See but Not to See.** *Brain* 1991, **114**:1523–1545.

A first thorough investigation of simultanagnosia using current cognitive theories and methods.

10. DAVIDOFF J, LANDIS T: **Recognition of Unfamiliar Faces in Prosopagnosia.** *Neuropsychologia* 1990, **28**:1143–1161.

11. •• HINTON GE, SHALLICE T: **Lesioning an Attractor Network: Investigations of Acquired Dyslexia.** *Psychol Rev* 1991, **98**:74–95.

A parallel distributed processing model of reading, which when lesioned, displays some of the characteristics of deep dyslexia.

12. BRUYER R: **Covert Face Recognition in Prosopagnosia: a Review.** *Brain Cogn* 1991, **15**:223–235.

13. • BURTON AM, YOUNG AW, BRUCE V, JOHNSTON RA, ELLIS AW: **Understanding Covert Recognition.** *Cognition* 1991, **39**:129–166.

A computer simulation of some aspects of covert face recognition in prosopagnosia.

14. •• MILNER AD, PERRETT DI, JOHNSTON RS, BENSON PJ, JORDAN TR, HEELEY DW, BETTUCI D, MORTARA F, MUTANI R, TERAZZI E *ET AL.*: **Perception and Action in 'Visual Form Agnosia'.** *Brain* 1991, **114**:405–428.

A striking dissociation between shape perception and awareness of thereof.

MJ Farah, Department of Psychology, Carnegie Mellon University, Pittsburgh, Pennsylvania 15213, USA.